

Demonstrating Immersive Gesture Exploration with GestureExplorer

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ABSTRACT

We demonstrate GestureExplorer, which features versatile immersive visualisations to grant the user free control over their perspective, allowing them to gain a better understanding of gestures. It provides multiple data visualisation views, and interactive features to support analysis and exploration of gesture data sets. This demonstration shows the potential of GestureExplorer for providing a useful and engaging experience for exploring gesture data.

Index Terms: H.5.1 [Multimedia Information Systems]: Artificial, augmented, and virtual realities;

1 INTRODUCTION

The design of intuitive gestures for new interactions has received much recent attention from researchers and gesture designers. Gesture Elicitation Studies (GES) are a popular way to elicit common gesture patterns that embody user preferences [6]. These studies have typically recorded elicited gestures in videos, which restricts their use to relatively small data sets given the time-consuming manual analysis they needed. Researchers have recently introduced tools that support the process of pattern analysis in GES through automation and visualisation [1, 2, 5]. However, the visualisation in these tools is implemented as a 2D projection, which obscures some information from the original 3D gesture data.

The emergence of Immersive Analytics (IA) provides new opportunities for gesture data analysis. IA exploits immersive environments to preserve the 3D nature of data to reduce cognitive effort of mentally manipulating 2D view.

In this paper, we present GestureExplorer, an immersive tool to facilitate the analysis and exploration of gesture data. GestureExplorer builds on prior work on gesture analysis tools by providing multiple interactive 3D gesture visualisations in a large virtual space to promote physical exploration. The demonstration reveals the potential of GestureExplorer for boosting gesture understanding and the process of exploration.

2 BACKGROUND

Existing studies preprocess the dataset by grouping it into small clusters, then visualise each cluster instead of each isolated data using: 1) node representations for the clusters and links connecting nodes for the representation of the relationship between them [2]. 2) a flat layout of all clustered data [1], in which a scatter plot or a density graph is drawn to provide an overview of the dataset. Some other studies attempt to overlap data together [4], which takes less space to display the entire dataset and meanwhile enables swift identification of agreements and differences among data. As for the visualisation of individual gesture data in a dataset, common approaches include: 1) drawing a static trajectory of that motion [3], and 2) picking frames of the gesture based on a fixed time interval,

then the poses of the gesture at the selected frames are visualised as small-multiple plots [2]. A variation to this is to link a series of pre-processed gesture poses to represent that motion [1]. 3) providing playback animation of the gesture data represented by a skeleton ontology [1, 2, 5]. Our tool allows users to choose either approach, and provides a novel feature combining all.

Though many approaches to visualise gesture data have been demonstrated in the previous work, no study has been done to investigate and evaluate the introduction of an immersive 3D environment into this field. Our tool is implemented in a virtual environment to provides users a free perspective to inspect the visualisations. Moreover, the immersive environment provides intuitive and embodied interactions with the visualisations, enabling an entertaining and engaging visual exploration as well as visual analysis of gesture data.

3 GESTUREEXPLORER: THE PROTOTYPE SYSTEM

In this demo, we will guide attendees through several features of GestureExplorer. These features include:

Gesture clustering and visualisation - GestureExplorer employs K-means clustering on gesture data sets, reducing manual efforts in identifying the common gesture pattern. It provides various visualisations for both gesture clusters and the individual gesture data within them - the former is visualised using a bubble metaphor, embedding an average gesture calculated from its member gestures while the latter is visualised as skeletons with motion trajectories drawn for each body joint (figure 1 left). However, our tool is not limited to full-body gestures, it could be potentially used for other kinds of gestures, too.

Cluster expansion, gesture animation and small multiples - In the virtual reality environment of GestureExplorer, users can move around gesture views and navigate the larger space using continuous movement via the controllers and select clusters or gestures using raycast. Users can unfold a cluster for its member gestures and inspect their animation or small multiples. Additionally, GestureExplorer offers users the option to control the playback of the animation along with a second animated view overlaid on small multiples that indicates the current time frame of the animation (figure 1 right).

Gesture arrangements, overview map and change cluster - Users can arrange gestures in different rationales for a better understanding of the relationship between gestures within and among clusters. There are currently 3 arrangements in our prototype - *global*, *local* and *line-up*. The *global* arrangement (figure 2 A) places gestures and clusters in relation to their consensus with the entire data set, while the *local* arrangement (figure 2 B) sets gestures around their respective clusters based on their similarity with the average gesture of that cluster. Under the *line-up* arrangement (figure 1 middle), gestures are sorted in a row, ordered by their local similarity. We offer a handheld *overview map* (figure 2) to facilitate users observation of outliers and similarity of gestures and clusters. Furthermore, users can move a gesture to another cluster where they think it belongs, which will consequently trigger an update of the placement of each gesture and cluster under current arrangement.

Trajectory stacking, motion heat map and trajectory filter - GestureExplorer also allows users to *stack gestures* (figure 3 A) for a fast identification of similarities and differences among the motion trajectories. Meanwhile, a *heat map* view (figure 3 B) can be applied to the stacked gestures to highlight the areas with the greatest amount

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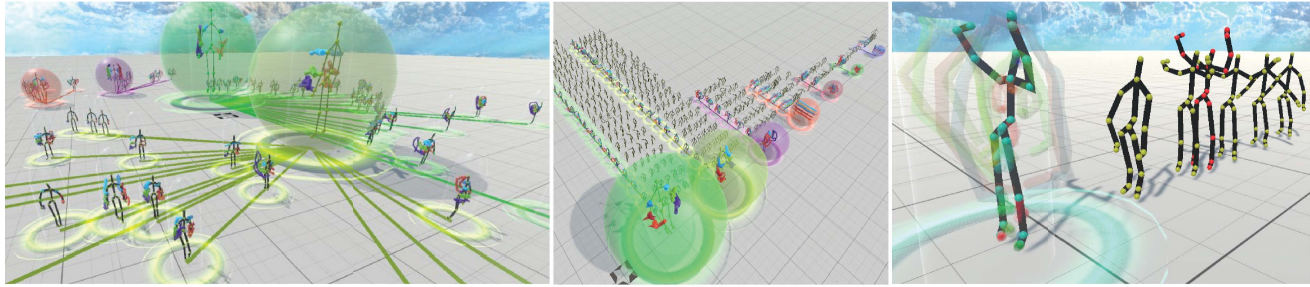


Figure 1: GestureExplorer supports immersive exploration of gesture data. Gestures are clustered by similarity and can be spatially arranged by similarity distance to each cluster (left), or in sorted order (middle). We provide several interactive features for exploring individual gestures such as trajectory visualisation, small multiples, and animation (right).

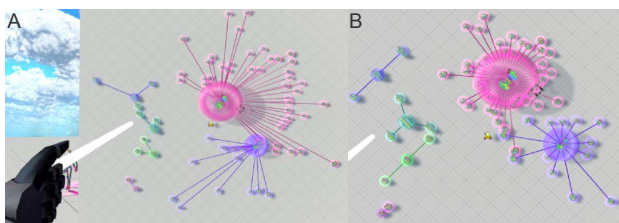


Figure 2: Handheld *overview map* displaying A) *Global* arrangement. Gestures are placed around the origin point (denoted by a white star on the floor). The distance between a gesture and the origin denotes the consensus of that gesture with the entire data set. B) *Local* arrangement. Gestures are placed around its assigned cluster. The distance between a gesture and its cluster denotes the within-cluster consensus of each gesture.

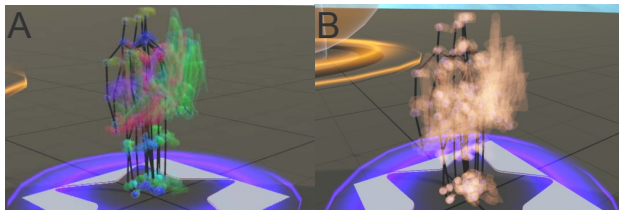


Figure 3: A) Trajectory stacking of several gestures. B) Heat map for the stacked trajectories. The more highlighted areas mean they are more common areas passed through by the gestures.

of motion using additive blending. To reduce visual occlusion, users can *filter out trajectories* they do not want to observe and focus only on interesting ones.

Our tool is implemented with an Oculus Rift S in Unity 3D version 2020.3.17f1. We implemented all features described above. We implemented an immersive menu, containing interactive buttons representing the described features. The menu stays in front of the user and can be shown or hidden by pressing the primary button on the controllers. To use the features in the menu, one first needs to activate the corresponding button in the menu, then select any valid object (e.g., clusters/gestures) and trigger the effect on it. Multiple features can be activated on an object at the same time. For instance, the user can observe the *trajectory stacking* view while using *trajectory filter* to isolate trajectories for a particular body part.

4 EXPECTED EXPERIENCE FROM DEMONSTRATION

We will make GestureExplorer available to download for people to try out. This has already been tested on different VR devices. Users will be able to gain an engaging and entertaining experience exploring and analysing gestures in the virtual reality environment of our prototype. This may raise awareness of the potential for future gesture analysing tools that incorporate immersive environments to enhance understanding and exploration of gestures. The implication of immersive data clustering and visualisation may impact beyond Gesture Elicitation Studies and may be useful to the application of Immersive Analytics for other data types.

5 CONCLUSION

We demonstrated GestureExplorer, an immersive system for exploring gesture data. Our prototype tool presents multiple immersive visualisations with interactive features to help users explore and gain a better understanding of gesture data sets. Our demonstration showed the potential value of such a tool.

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